

Question Number	Answer	Mark
2(a)(i)	The only correct answer is D 1, 2 and 3 <i>A is not correct because all three processes occur in interphase</i> <i>B is not correct because all three processes occur in interphase</i> <i>C is not correct because all three processes occur in interphase</i>	(1)

Question Number	Answer	Mark
2(a)(ii)	The only correct answer is C (V Z X Y W) <i>A is not correct because prophase occurs after interphase and before metaphase</i> <i>B is not correct because prophase occurs after interphase and before metaphase</i> <i>D is not correct because metaphase occurs after prophase</i>	(1)

Question Number	Answer	Additional guidance	Mark
2(a)(iii)	(×)750	ACCEPT 0.5 mm tolerance ACCEPT answer between the range: (×)729 to 771 REJECT answers with a unit	(1)

Question Number	Answer	Additional guidance	Mark
2 (b)	• correct total number of cells (1) • correct number of cells in interphase (1)	<u>Example of calculation:</u> $450 \div 0.375$ $1200 - 450 = 750$ only ecf is an incorrectly calculated number minus 450 shown in working Correct answer with no working shown scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
2(c)	A description that makes reference to two of the following points: • {chromosomes / chromatin / DNA / chromatid} condense (1) • nuclear {envelope / membrane} breaks down (1) • centrioles {move to poles of the cell / produce spindle fibres / produce microtubules} (1)	ACCEPT {DNA / chromatin} coils around histones ACCEPT chromatin starts forming chromosomes ACCEPT nuclear {envelope / membrane} disappears / nucleolus disappears ACCEPT spindle fibres form without ref to centrioles ACCEPT centrosome for centriole REJECT centromere for centriole	(2)

Question Number	Answer	Additional guidance	Mark
2(d)(i)	correct example of a plant tissue (1)	mark first answer e.g. xylem, phloem, meristem, sclerenchyma, parenchyma, spongy mesophyll, palisade mesophyll, endosperm REJECT cells e.g. palisade mesophyll cell IGNORE root / root hair	(1)

Question Number	Answer	Additional guidance	Mark
2(d)(ii)	asexual reproduction (1)	ACCEPT some stages from the formation of plant gametes ACCEPT cloning / reproduction of single celled organisms ACCEPT replacement of cells IGNORE asexual unqualified IGNORE replace tissues	(1)